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PT5053 – Evidence Based Practice

Critical Appraisal of the Efficacy of heel lifts versus calf muscle eccentric exercise for mid-portion Achilles tendinopathy (HEALTHY): a randomised trial

Word Count: 775

Introduction

Evidence based practice in physiotherapy requires physiotherapists to appraise research critically to determine whether the findings are trustworthy and appropriate for practice. To apply this, a randomised controlled trial (RCT) by Rabusin et al in 2021, which compared heel lifts with eccentric calf muscle exercises for mid portion Achilles tendinopathy, has been critically appraised using the CASP RCT checklist. This examines the study's methodological quality, such as design, sampling, data collection, analysis, strengths, limitations and clinical relevance. As Achilles tendinopathy is prevalent in physiotherapy, evaluating the effectiveness of commonly prescribed interventions is important for deciding interventions.

Study Design

The study was a parallel group randomised superiority trial, which is a rigorous design or evaluating intervention effectiveness. The study followed the CONSORT guidelines, enhancing the transparency of the study. The study was randomised through a computer using permuted blocks, and allocation concealment was maintained by an independent online service. This aligns with the CASP as it minimises selection bias. Blinding of participants and research staff (CLR and JMG) was not feasible due to the nature of the treatments (Rabusin *et al.*, 2020). However, research staff analysing the data (CLR, HBM and JMG) were blinded to group allocations (Rabusin *et al.*, 2020). This reduces the chances of detection bias.

Sampling and Recruitment

100 adults with diagnosed mid portion Achilles tendinopathy were recruited through advertisement. The inclusion and exclusion criteria were clear and appropriate; however, the sampling may introduce selection bias as the sample represented many patients with chronic symptoms. The median duration found in Table 1 was between 18 and 22 ½ months (Rabusin *et al.*, 2020). This limits the study's applicability to the acute diagnosis of Achilles tendinopathy. The characteristics of the sample group such as age, sex, height and other baseline characteristics, were balanced between the groups. This supports the validity of the study.

Data Collection Method

There were multiple data collection methods that related to physiotherapy. The primary outcome measure was the VISA-A questionnaire, which is a measure of pain, function and activity. Secondary outcome measures included ultrasound, tendon thickness, visual analogue scale (VAS) for pain severity, EQ 5D 5L for quality

of life, physical activity, calf muscle endurance and patient reported improvement (PIC). These measures captured subjective and objective aspects of tendon health.

Data was collected at the start, 2nd, 6th and 12th week to see the change over time. A limitation to self reported adherence is that patients may have social desirability bias. They may overreport positive symptoms and underreport negative ones to present themselves better. The 12 week follow up limits insight into long term outcomes, which is relevant to Achilles tendinopathy management.

Data Analysis and Findings

The data analysis aligned with the CASP criteria. To preserve the benefits of randomisation, Rabusin *et al.* used an intention to treat analysis and applied multiple imputation to address the missing data. This reduces the chances of attrition bias. ANCOVA was used to compare the continuous outcome measures while adjusting for the baseline differences. Relative risk and number needed to treat (NNT) were used for binary outcomes.

The VISA-A showed a significant improvement favouring heel lifts as the adjusted mean was 9.6, $p = 0.016$ (Rabusin *et al.*, 2020). It missed the 10 points, the predetermined minimum important difference, which questions the clinical significance of the study. Secondary outcome measures showed more reductions in pain and small improvements in quality of life in the heel lift group, but tendon structure and calf function showed no difference between the groups. Adherence was better in the heel lift group which may have contributed to better outcomes.

Strengths, Limitation and Trustworthiness

The strengths of the study include relevant outcome measures, blinded data analysts, robust randomisation and allocation concealment. The limitations include the lack of participant blinding, short follow ups, self reported adherence, which potentially leads to selection bias. But the study demonstrates good internal validity and moderate external validity. However, the confidence in the findings indicates good evidence but possible uncertainties.

Relevance To Physiotherapy

Heel lifts have demonstrated that they can be better than short term improvements to pain and function with more adherence and less risk. They may serve as an intervention for patients experiencing exercise compliance compared to eccentric calf exercises. However, eccentric calf exercises remain a vital intervention due to the long term benefits. As there aren't long term follow ups in the study, it limits the conclusions about the outcomes, which are significant in Achilles tendinopathy management.

Conclusion

This RCT provides credible evidence that heel lifts offer short term benefits over eccentric exercise for Achilles tendinopathy. There are limitations, however, the findings show that adding heel lifts promotes adherence alongside progressive loading.

Appendix

Critical Appraisal Skills Programme

CASP Checklist:

For Randomised Controlled Trials (RCTs)

Reviewer Name:	
Paper Title:	Efficacy of heel lifts versus calf muscle eccentric exercise for mid- portion Achilles tendinopathy (HEALTHY): a randomised trial
Author:	Rabusin, C. L., Menz, H. B., McClelland, J. A., Evans, A. M., Malliaras, P., Docking, S. I., Landorf, K. B., Gerrard, J. M. and Munteanu, S. E.
Web Link:	https://pubmed.ncbi.nlm.nih.gov/32988930/
Appraisal Date:	10/3/2026

During critical appraisal, never make assumptions about what the researchers have done. If it is not possible to tell, use the "Can't tell" response box. If you can't tell, at best it means the researchers have not been explicit or transparent, but at worst it could mean the researchers have not undertaken a particular task or process. Once you've finished the critical appraisal, if there are a large number of "Can't tell" responses, consider whether the findings of the study are trustworthy and interpret the results with caution.

Section A Is the basic study design valid for a randomised controlled trial?	
1. Did the study address a clearly formulated research question? The study defines its aim to compare the efficacy of in shoe heel lifts to calf muscle eccentric exercise in reducing pain and improving function for Achilles tendinopathy Population studied is adults with mid portion achilles tendinopathy Intervention is heel lifts Comparator chosen is eccentric exercise Outcome measures include pain, function and VISA-A	☒ Yes ☐ No ☐ Can't Tell
CONSIDER: *Was the study designed to assess the outcomes of an intervention?* *Is the research question 'formulated' in terms of:* • *Population studied* • *Intervention given* • *Comparator chosen* • *Outcomes measured?*	

2. Was the assignment of participants to interventions randomised? The study shows that "Randomisation was provided by an independent computer (online) generated service, using permuted blocks of random sizes of 2,4,6". "The allocation was provided once eligibility was confirmed and baseline measures were completed." This shows that the allocation concealment was ensured.	☒Yes ☐ No ☐ Can't Tell
CONSIDER: • *How was randomisation carried out? Was the method appropriate?* • *Was randomisation sufficient to eliminate systematic bias?* • *Was the allocation sequence concealed from investigators and participants?*	
3. Were all participants who entered the study accounted for at its conclusion? The CONSORT flow diagram shows 100 randomised, 80 completed 12-week follow-up (40 per group) Losses to follow up were documented. "Unable to contacted (n=10)" in each group. The study used intention to treat analysis and multiple imputation	☒Yes ☐ No ☐ Can't Tell
CONSIDER: • *Were losses to follow-up and exclusions after randomisation accounted for?* • *Were participants analysed in the study groups to which they were randomised (intention-to-treat analysis)?* • *Was the study stopped early? If so, what was the reason?*	
Section B Was the study methodologically sound?	
4. (a) Were the participants 'blind' to intervention they were given? "Due to the nature of the interventions, the participants were blinded." This is because heel lifts and eccentric exercises are visibly different.	☐Yes ☒ No ☐ Can't Tell

(b) Were the investigators 'blind' to the intervention they were giving to participants? "Research staff administering the interventions were not blinded." This is because the intervention providers could not be blinded.	☐ Yes ☒ No ☐ Can't Tell
(c) Were the people assessing/analysing outcome/s 'blinded'? The study says that "research staff analysing data (CLR, HBM and JMG) were blinded to the group allocation (masking used).	☒ Yes ☐ No ☐ Can't Tell
5. Were the study groups similar at the start of the randomised controlled trial? Table 1 shows no relevant baseline differences: "There were no clinically relevant differences in baseline characteristics between groups." Variables such as age, sex, BMI, symptom duration and VISA-A scores were comparable	☒ Yes ☐ No ☐ Can't Tell
CONSIDER: • *Were the baseline characteristics of each study group (e.g. age, sex, socio-economic group) clearly set out?* • *Were there any differences between the study groups that could affect the outcome/s?*	
6. Apart from the experimental intervention, did each study group receive the same level of care (that is, were they treated equally)? Both groups received the same pain monitoring activity programme, follow up schedule and outcome assessments. The study says that "both groups received a handout explaining their allocated intervention and a modified physical activity programme on the pain monitoring model."	☒ Yes ☐ No ☐ Can't Tell
CONSIDER:	

• *Was there a clearly defined study protocol?* • *If any additional interventions were given (e.g. tests or treatments), were they similar between the study groups?* • *Were the follow-up intervals the same for each study group?*	
Section C: What are the results?	
7. Were the effects of intervention reported comprehensively? A power calculation was taken. "An a priori sample size calculation estimated that a total of 92 participants (46 per group) would be needed. This shows the appropriate planning to ensure adequate statistical power. The primary outcome was defined. "The primary outcome was the VISA-A". Secondary outcomes were also listed such as ultrasound tissue characterisation, pain severity (VAS), EQ-5D-5L, physical activity, calf muscle function, patient impression change, adverse events and adherence. All the outcome measures were described and justified. Continuous outcomes were expressed using means, standard deviations, adjusted mean differences and 95% confidence intervals (CIs) Binary outcomes such as PIC improvement were expressed using "Relative risk, risk difference and number needed to treat" Table 2 and the text report results at baseline, 2 weeks, 6 weeks and 12 weeks for both groups. "The mean VISA- A score improved by 26.0 points (95% CI 19.6 to 32.4) in the heel lifts group and by 17.4 points (95% CI 9.5 to 25.3) in the eccentric exercise group." This allows clear comparison across the time points. There was missing data, the follow up was 80% at 12 weeks so the authors used "Multiple imputation was used to replace missing data using five iterations, with sex, age, baseline scores and group allocation used as predictors,38 for all primary and secondary outcome measures." This reduces bias from the missing data. There wasn't a differential drop out between the study groups that could affect the results are both groups had identical follow up numbers. "40 per group" completed the 12 week assessment. This reduces the risk of attrition bias.	☒ Yes ☐ No ☐ Can't Tell

The authors acknowledged the potential biases identified such as lack of participant and provider blinding, self reported adhereance and short follow up duration. However the fact that "neither the participants, nor research staff administering the interventions (CLR and JMG), assessing outcomes (CLR), or entering data (CLR) were blinded," strengthens the trustworthiness of the study. The statistical tests used were ANCOVA for continuous, relative risk, risk difference, NNT for binary outcomes and multiple imputation for missing data. P values were reported for the primary outcome measure being "p = 0.016." The P values were reported for all key comparisons.	
CONSIDER: • *Was a power calculation undertaken?* • *What outcomes were measured, and were they clearly specified?* • *How were the results expressed? For binary outcomes, were relative and absolute effects reported?* • *Were the results reported for each outcome in each study group at each follow-up interval?* • *Was there any missing or incomplete data?* • *Was there differential drop-out between the study groups that could affect the results?* • *Were potential sources of bias identified?* • *Which statistical tests were used?* • *Were p values reported?*	
8. Was the precision of the estimate of the intervention or treatment effect reported? Confidence intervals were provided for all key outcomes "95% CI 1.8 to 17.4". This shows the precision and clinical relevance of the results.	☒Yes ☐ No ☐ Can't Tell
CONSIDER: • *Were confidence intervals (CIs) reported?*	
9. Do the benefits of the experimental intervention outweigh the harms and costs? Heel lifts produced greater improvement in VISA-A and pain. Adverse events were minimal and similar between groups. Heel lifts are inexpensive and don't require training. The	☒Yes ☐ No ☐ Can't Tell

study states that "participants in the heel lifts group were more likely to report an improvement in pain (RR 1.4, 95% CI 1.0 to 2.0, NNT 4) and function (RR 1.6, 95% CI 1.0 to 2.3, NNT 4), as measured using the PIC." No cost effectiveness analysis was performed.	
CONSIDER: • *What was the size of the intervention or treatment effect?* • *Were harms or unintended effects reported for each study group?* • *Was a cost-effectiveness analysis undertaken? (Cost-effectiveness analysis allows a comparison to be made between different interventions used in the care of the same condition or problem.)*	
Section D: Will the results help locally?	
10. Can the results be applied to your local population/in your context? The participants are adults with diagnosed mid portion Achilles tendinopathy which is a common condition in physiotherapy settings. The study describes the participants as "aged at least 18 years of age, with clinically diagnosed and ultrasonographically confirmed mid-portion Achilles tendinopathy." This matches the typical physiotherapy caseload found in musculoskeletal physiotherapy. There may be possible differences between the patient population in physiotherapy and the study participants. The sample had a symptom duration median of 18-22 ½ months, indicating that the participants had chronic Achilles tendinopathy. If the population includes acute cases, treatment will differ. The outcomes are important for people suffering from chronic Achilles tendinopathy as the outcome measures measured pain, function, tendon structure, quality of life and calf muscle performance. The measures are relevant to physiotherapy practice to maintain a holistic approach to treatment. The VISA-A reflects the functional capacity and symptom severity which link clinical reasoning and decision making of the intervention. The outcomes I would have wanted information on that weren't reported were the long term follow up after 12 weeks, return to sport or	☒ Yes ☐ No ☐ Can't Tell

activity outcomes, cost effectiveness and patient satisfaction beyond the PIC scale These outcomes would be important information for physiotherapists planning long term management. Key limitations were the lack of participant and provider blinding, self reported adherence, short follow up period and recruitment via advertisement, which could lead to selection bias. The study states that "neither the participants, nor research staff administering the interventions (CLR and JMG), assessing outcomes (CLR), or entering data (CLR) were blinded." So these findings should be considered when applying the findings to practice.	
CONSIDER: • *Are the study participants similar to the people in your care?* • *Would any differences between your population and the study participants alter the outcomes reported in the study?* • *Are the outcomes important to your population?* • *Are there any outcomes you would have wanted information on that have not been studied or reported?* • *Are there any limitations of the study that would affect your decision?*	
11. Would the experimental intervention provide greater value to the people in your care than any of the existing interventions? The intervention, heel lifts, provides a greater value than existing options as they are low cost, require no equipment or training, have higher adherence and show similar or better short term outcomes than eccentric exercises. The study concludes that "Healthcare providers can consider heel lifts as a first- line treatment for mid- portion Achilles tendinopathy." It supports the use of heel lifts for mid portion Achilles tendinopathy in physiotherapy.	☒ Yes ☐ No ☐ Can't Tell
CONSIDER: • *What resources are needed to introduce this intervention taking into account time, finances, and skills development or training needs?* • *Are you able to disinvest resources in one or more existing interventions in order to be able to re-invest in the new intervention?*	

APPRAISAL SUMMARY: *List key points from your critical appraisal that need to be considered when assessing the validity of the results and their usefulness in decision-making.*		
Positive/Methodologically sound	**Negative/Relatively poor methodology**	**Unknowns**
Clear research question **Blinded data analysts** **Validated outcome measures** **Comprehensive reporting** **High adherence in heel lift group**	**No participant blinding** **Self reported adherence** **Short follow up** **Recruitment with advertisement leading to potential selection bias**	**Long term effectiveness** **Cost effectiveness** **Impact on acute tendinopathy cases**

Referencing recommendation:

CASP recommends using the Harvard style referencing, which is an author/date method. Sources are cited within the body of your assignment by giving the name of the author(s) followed by the date of publication. All other details about the publication are given in the list of references or bibliography at the end.

Example:

Critical Appraisal Skills Programme (2024). CASP (insert name of checklist i.e. randomised controlled trials (RCTs) Checklist.) [online] Available at: insert URL. Accessed: insert date accessed.

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Reference List

Rabusin, C. L., Menz, H. B., McClelland, J. A., Evans, A. M., Malliaras, P., Docking, S. I., Landorf, K. B., Gerrard, J. M. and Munteanu, S. E. (2020) 'Efficacy of heel lifts versus calf muscle eccentric exercise for mid-portion Achilles tendinopathy (HEALTHY): a randomised trial', *British Journal of Sports Medicine*, 55(9), p. bjsports-2019-101776. doi: https://doi.org/10.1136/bjsports-2019-101776.

Critical Appraisal Skills Programme (2024) *CASP Randomised Controlled Trials Checklist*. [online] Available at: https://casp-uk.net/casp-tools-checklists/ [Accessed 13 March 2026].